

Douglas College



Module Code:

CSIS 4495 – 071

Module Title:

Applied Research Project

Year & Semester:

2025 Fall

Progress Report-0035

Alternative Industries: Weather Driver

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Instructor: Bambang, Sarif

Due Date: Nov 22nd, 2025

Date	Hours	Description of work Done
Nov 9	4	Architecture Planning: Completed requirements review and detailed planning for the upcoming Authentication and Maps workflow implementations. This defined the foundational architecture for the features.
Nov 11	5	Maps Flow Implementation: Successfully built the complete Maps Flow, integrating geolocation, stop selection, and the core navigation logic for map-based routing within the Weather Driver application.
Nov 15	3	AWS Cognito Integration: Implemented AWS Cognito Login Redirection, establishing a secure authentication pathway and ensuring compatibility with the app's internal navigation system.
Nov 16	2	System Verification & Setup: Performed environment cleanup and rigorous testing of both AWS Cognito redirection and Google Maps functionality, ensuring consistent and proper local configuration across the development team's systems.
Nov 18	6	Build Stabilization & Structure: Performed significant project restructuring and added Authorizer & Permissions logic. Prepared the building environment for EAS Integration and conducted necessary adjustments to stabilize the core authentication system.
Nov 19	3	Google OAuth Integration: Migrated the build configuration to the server to facilitate Google Sign-In Integration, ensuring seamless compatibility between the mobile frontend and backend OAuth settings.

Date	Hours	Description of work Done
Nov 20	4	Frontend & Auth Logic: Completed the Signup UI (with Google button) and the Login Screen. Implemented validation logic and refined the overall authentication flow.
Nov 20	3	Client Sync & Key Issues: Attended a critical client meeting to highlight the missing SHA-1 fingerprint on the Google Console and non-functional client ID/secret. Also requested the necessary Weather API keys needed for future feature implementation.
Nov 22	6	Final Security Configuration: Attended a follow-up client meeting to adjust the SHA-1 fingerprint and successfully synchronized AWS Cognito with the client's Web and App Clients, integrating the necessary key and secret. Updated the Manual Login/Signup flows and assisted group members with the EAS build process and APK deployment.

Summary of Work:

This period was highly productive, focusing intensely on establishing the project's most critical pillars: secure authentication and functional maps navigation flow. My work began with the essential architectural setup, specifically integrating the AWS Cognito User Pool. This involved configuring OAuth scopes and callback URLs and aggressively troubleshooting complex configuration mismatches and domain issues that were initially hindering a secure login pipeline. Parallely, I immediately addressed a major blocker with our geospatial features, successfully migrating from deprecated Google Maps APIs to the modern Maps and Routes APIs, which restored vital functionalities like location search and route preview.

A key part of the success involved bridging technical gaps with the client. On November 20th, I conducted a demonstration highlighting the critical missing SHA-1 fingerprint and non-functional client OAuth keys, which was instrumental in pushing forward necessary security updates. This led directly to a follow-up meeting on November 22nd where I worked collaboratively adjusting the SHA-1 fingerprint on the Google Console and synchronizing AWS Cognito with the

client's new, functioning keys. This essential configuration work stabilized the entire security perimeter and enabled the full deployment of the Signup and Login screens with robust Google Sign-In support.

Finally, I delivered the end-to-end Maps Flow, integrating geolocation, stop selection, and routing logic, thus completing the primary navigation features for the application. By resolving major technical configurations, integrating core security systems, and actively managing client-side key issues, this phase has moved the project from architecture to a fully functional, secure foundation, positioning us optimally for the next stage of feature development.